

MICHAEL BOGALE

Artificial Intelligence | Data Scientist

[Portfolio Website](#) | [LinkedIn](#) | [GitHub](#)

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PROFESSIONAL OBJECTIVE

A highly motivated MSc graduate in Artificial Intelligence (with distinction) equipped with a solid foundation in data science, machine learning, deep learning, and NLP. I am seeking a challenging role in data science and AI to apply my skills in Python, SQL, with hands-on experience with advanced analytics to drive data-driven solutions. I am passionate about leveraging my technical expertise to solve real-world problems and contribute to organisational growth.

EDUCATION

MSc in Artificial Intelligence

SCQF Level 11

University of Stirling, UK (September 2022 – November 2023)

Graduated with Distinction

- Relevant Courses: Statistics for Data Science, Machine Learning, Deep Learning for Computer Vision and NLP, Stochastic Processes, Optimisation, Manipulating Data

Google Advanced Data Analytics Professional Certification

Coursera (August 2024 – Present)

- Gaining advanced skills in applying statistical analysis, machine learning, and data visualisation techniques to solve business problems
- Topics covered: Data Wrangling, Exploratory Data Analysis (EDA), Hypothesis Testing, Regression Analysis, Predictive Modeling, and Optimization
- Hands-on projects: Utilising SQL and Python for large-scale data processing, advanced querying, and model deployment using TensorFlow and Scikit-learn
- Focused on real-world case studies involving advanced analytics, data-driven decision-making, and scaling AI models for business use cases

Boot Camp in Data Analytics

Masterschool (Online) (January 2023 – Present)

- Relevant Courses: Business Analytics using SQL, Data Storytelling

SKILLS

Programming Languages

- **Python (Advanced):** Experience in AI/ML, data analysis, automation, and Raspberry Pi projects. Libraries: Pandas, NumPy, TensorFlow, PyTorch, Scikit-learn, NLTK
- **R:** Statistical computing
- **SQL:** Queries, large dataset handling
- **Bash/Shell Scripting:** Automating tasks and managing Raspberry Pi environments in Linux

Technologies

- **TensorFlow & PyTorch:** Developing and optimising deep learning models
- **Scikit-learn:** Implementing classical ML algorithms, feature engineering, and model evaluation
- **NLTK:** Text analysis and natural language processing
- **Pandas & NumPy:** Data manipulation, preprocessing, numerical analysis
- **Linux:** Command-line operations, scripting, task automation, Raspberry Pi shell scripting

Tools

- **Jupyter Notebook & Google Colab:** Interactive environments for coding and experimentation
- **Git & GitHub:** Version control and collaborative development
- **Excel & Spreadsheets:** Data analysis, reporting, and visualisation
- **Raspberry Pi:** Prototyping, automation, and IoT projects

Core Skills

- **Machine Learning & Deep Learning:** Model training, optimization, evaluation
- **NLP & Text Analysis:** Data cleaning, sentiment analysis, feature extraction
- **Statistical Analysis:** Hypothesis testing, regression
- **Data Visualization:** Using Matplotlib, Seaborn for insights
- **Problem Solving & Algorithm Development:** Tackling complex technical challenges
- **Teamwork & Communication:** Collaborative project execution, clear technical communication

PROJECTS

Dissertation: Predicting the Impact of Road Conditions on Electric Vehicles Using AI Algorithms

- Analysed large **EV sensor datasets from Kaggle** to predict vehicle performance under varying road conditions
- Employed **Genetic Algorithm (DEAP library)** for optimization
- Utilised **Keras** for model development and machine learning tasks
- Integrated **Google API** for real-time route and road condition data

Deep Learning Model for Satellite Image Classification

- Built a dataset of **2,400 manually labelled images** from Google Earth
- Applied **data augmentation** techniques to enhance image diversity
- Tested and evaluated **CNN architectures** based on past performance in image classification tasks, focusing on cars and trees

Natural Language Processing on 1,176 Books from Project Gutenberg

- Performed extensive **data cleaning** and **metadata extraction**
- Conducted **sentiment analysis** on selected books
- Executed **exploratory and descriptive data analysis** to extract insights from text data

CERTIFICATIONS

- Google Cloud: [Introduction to Generative AI \(Coursera\)](#)
- Google Cloud: Advanced Data Analytics Professional Certification (In Progress)
- Sololearn: [Python for Data Science](#)
- Sololearn: [Machine Learning](#)
- Sololearn: [Python Developer](#)
- Datacamp: Python
- Datacamp: [SQL](#)
- Datacamp: [Data Analysis in Spreadsheets](#)
- Udemy: [MS Excel for Data Analysis](#)

AWARDS

- [University of Stirling Academic Award Letter](#)